

Advanced Statistical Methods II

Lecture 14: LAD Regression

Monitirtha Dey



BSDS Third Year Students
Indian Statistical Institute

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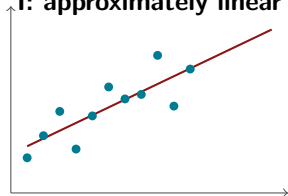
He used this procedure while trying to reconcile incoherent measures that were used to estimate the shape of the earth.

The easy calculus of the least-squares method made least squares much more popular than the LAD method.

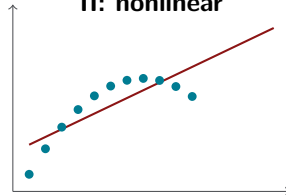
Anscombe's Quartet



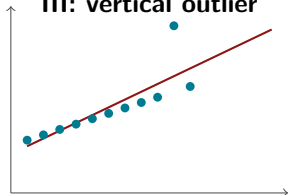
I: approximately linear



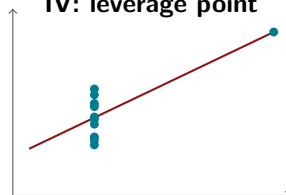
II: nonlinear



III: vertical outlier



IV: leverage point



All four have $\bar{x} = 9$, $\bar{y} = 7.50$, $r \approx 0.816$, and OLS line $\hat{y} \approx 3 + 0.5x$.

Source: Anscombe (1973), *The American Statistician*, 27(1), 17–21.

Identical summaries, different data.

What LAD changes in the quartet

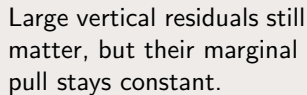


Dataset	OLS slope	LAD slope
I	0.500	0.480
II	0.500	0.500
III	0.500	0.345
IV	0.500	0.496

Dataset III shows LAD's reduced sensitivity to an extreme response.

Dataset IV shows the limitation: one high-leverage observation can control both OLS and LAD.

Plots reveal model failures that summary statistics and loss values can hide.



Real data 1: industrial stack loss



Brownlee recorded 21 days of ammonia-oxidation plant operation (oxidizing ammonia to nitric acid). Response: ammonia lost from the stack.
Predictor: air flow, cooling-water temperature, and acid concentration.

Fit on all 21 days	MAE	RMSE
OLS	2.37	2.92
LAD	2.00	3.29

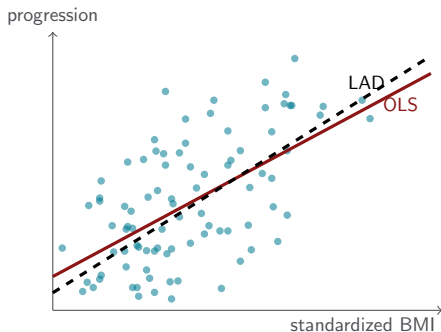
Air-flow coefficient

	All days	Remove day 1
OLS	0.716	0.662
LAD	0.832	0.830

This comparison illustrates sensitivity of the two methods.

Source: Brownlee (1965), *Statistical Theory and Methodology in Science and Engineering*; calculations in the accompanying source.

Real data 2: diabetes progression and BMI



The 442-patient data show broad vertical spread at similar BMI values.

	MAE	RMSE
OLS	51.80	62.37
LAD	51.15	63.09

The preferred fit depends on whether the target is a conditional mean or median.

Source: Efron et al. (2004), *Annals of Statistics* 32(2); every fifth point shown, fits use all observations.

Conditional mean

OLS estimates $\mathbb{E}(Y | X = x)$ when the linear mean model is correct.

Best when the decision cost is quadratic or the mean is scientifically central.

Conditional median

LAD estimates $\text{median}(Y | X = x)$ when the linear median model is correct.

Best when the decision cost is absolute or the typical response is central.

For observations (y_i, x_i) with $x_i \in \mathbb{R}^p$:

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n (y_i - x_i^\top \beta)^2$$

smooth, strictly convex under full
rank

$$\hat{\beta}_{\text{LAD}} = \arg \min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n |y_i - x_i^\top \beta|$$

convex, piecewise linear, nonsmooth
at zero

Same linear predictor, different target and geometry.

The median minimizes absolute loss



Let a be a constant prediction and Y have distribution function F . Consider

$$g(a) = \mathbb{E}|Y - a|.$$

Any a satisfying

$$F(a^-) \leq \frac{1}{2} \leq F(a)$$

minimizes $g(a)$. These are precisely the medians.

Conditional on $X = x$, the same argument makes the LAD target the conditional median.

$$Q_{0.5}(Y \mid X = x) = x^\top \beta_0,$$
$$\hat{\beta}_{\text{LAD}} = \arg \min_{\beta} \sum_i \rho_{0.5}(y_i - x_i^\top \beta), \quad \rho_{0.5}(u) = \frac{1}{2}|u|.$$

Interpretation

Holding other predictors fixed, a one-unit increase in x_j shifts the modeled conditional median by β_j .

Quantile regression extends LAD



For $0 < \tau < 1$, define the check loss

$$\rho_{\tau}(u) = u\{\tau - \mathbf{1}(u < 0)\}.$$

Then

$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_i \rho_{\tau}(y_i - \mathbf{x}_i^{\top} \beta).$$

Koenker and Bassett (1978) replaced a single conditional-mean line with a family

$$Q_{\tau}(Y | X = x) = x^{\top} \beta(\tau), \quad 0 < \tau < 1.$$

Their construction yields several important results:

- ▶ minimizing check loss estimates the conditional τ -quantile without specifying a full error distribution.
- ▶ the subgradient condition balances predictor-weighted residual signs at level τ .
- ▶ the estimator is equivariant to shifts and positive rescaling of the response. (Check!)
- ▶ LAD can be formulated as a linear program, allowing standard optimization softwares to find a global optimum reliably*.

* This does not necessarily mean that the computation is trivial. It means the optimization problem has a well-understood structure and does not have the local-minimum difficulties found in many nonconvex models.

- Full column rank does not guarantee coercivity. Under suitable general-position/nondegeneracy conditions, the LAD solution is typically unique; nevertheless, uniqueness is not guaranteed for every realized dataset.

$$Q(\beta) = \sum_{i=1}^n |r_i(\beta)| = \sum_{i=1}^n |y_i - \mathbf{x}_i^\top \beta|.$$

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For a convex function $f : \mathbb{R}^p \rightarrow \mathbb{R}$, a vector $g \in \mathbb{R}^p$ is called a **subgradient** of f at x_0 if

$$f(x) \geq f(x_0) + g^\top (x - x_0) \quad \text{for every } x \in \mathbb{R}^p.$$

The collection of all subgradients of f at x_0 is called the *subdifferential* and is denoted by

$$\partial f(x_0).$$

If f is differentiable at x_0 , then the subgradient reduces to the usual gradient:

The basic example for LAD is

For $r > 0$,

while for $r < 0$,

At $r = 0$, the ordinary derivative does not exist because the left derivative is -1 and the right derivative is 1 . However, every slope between these two values defines a supporting line. Hence

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$$\partial|r| = \begin{cases} \{1\}, & r > 0, \\ [-1, 1], & r = 0, \\ \{-1\}, & r < 0. \end{cases}$$
$$s_i \in \partial |r_i|,$$
$$s_i = \begin{cases} 1, & r_i > 0, \\ \text{any value in } [-1, 1], & r_i = 0, \\ -1, & r_i < 0. \end{cases}$$

Subgradients for LAD

Since

$$r_i(\beta) = y_i - \mathbf{x}_i^\top \beta,$$

we have

$$\frac{\partial r_i(\beta)}{\partial \beta} = -x_i.$$

For $r_i \neq 0$, the contribution of observation i to the gradient is $-x_i \text{sign}(r_i)$.

More generally, allowing for zero residuals, the subdifferential of Q is

$$\partial Q(\beta) = \left\{ -\sum_{i=1}^n x_i s_i : s_i \in \partial |r_i| \right\}.$$

Writing

$$s = (s_1, \dots, s_n)^\top,$$

this becomes

$$\partial Q(\beta) = \{-X^\top s : s_i \in \partial|r_i|\}.$$

Subgradients for LAD



A fundamental result in convex optimization is

$$\hat{\beta} \text{ minimizes } Q \iff 0 \in \partial Q(\hat{\beta}).$$

For LAD, this means that $\hat{\beta}$ is optimal if and only if there exists a valid choice of $s_1, \dots, s_n$ satisfying

$$s_i \in \partial |r_i(\hat{\beta})|$$

such that

$$0 = -X^T s.$$

Equivalently,

$$X^T s = 0.$$

Thus the LAD optimum satisfies a balance condition:

$$\sum_{i=1}^n x_i s_i = 0.$$

For observations with nonzero residuals, s_i is simply the sign of the residual. For observations with zero residuals, s_i may take any value between -1 and 1 , which allows the balance condition to hold.

$$\frac{\partial |r_i|}{\partial r_i} = \text{sign}(r_i) = \begin{cases} 1, & r_i > 0, \\ -1, & r_i < 0. \end{cases}$$
$$|\text{sign}(r_i)| = 1 \quad \text{whenever } r_i \neq 0.$$
$$r_i = 0.01 \quad \text{and} \quad r_j = 100$$
$$\text{sign}(0.01) = \text{sign}(100) = 1.$$

(reason behind being less sensitive to extreme response values)

Corresponding derivation for OLS



More precisely, in LAD, observation i contributes $-x_i \text{sign}(r_i)$ when $r_i \neq 0$.
For OLS,

$$Q_{\text{OLS}}(\beta) = \sum_{i=1}^n r_i(\beta)^2.$$

Differentiating,

$$\nabla Q_{\text{OLS}}(\beta) = -2 \sum_{i=1}^n x_i r_i.$$

Thus the contribution of observation i is

$$-2x_i r_i.$$

Contribution *grows linearly* with the size of the residual.

For instance, residuals 0.01 and 100 have residual-dependent magnitudes

$$0.01 \quad \text{and} \quad 100,$$

respectively. Hence the latter contributes 10,000 times as much through the residual term.

The LAD optimality condition

$$X^\top s = 0$$

can be written as

$$\sum_{i=1}^n x_i s_i = 0.$$

Hence the covariate vectors corresponding to positive and negative residuals must balance, with observations having zero residuals allowed to contribute fractional weights between -1 and 1 .

If $r_i > 0$, then

$$s_j = 1,$$

while if $r_i < 0$, then

$$s_j = -1.$$

If $r_j = 0$, then

$$s_j \in [-1, 1].$$

Thus the zero-residual observations provide exactly the flexibility needed for the subgradient balance condition to hold.

Intercept-only Model

Consider the intercept-only model

$$y_i = \beta + \varepsilon_i,$$

so that

$$x_i = 1 \quad \text{for every } i.$$

Then the LAD objective is

$$Q(\beta) = \sum_{i=1}^n |y_i - \beta|,$$

and the optimality condition becomes

$$\sum_{i=1}^n s_i = 0.$$

This says, roughly, that the numbers of positive and negative residuals must balance, with observations satisfying $y_i = \beta$ supplying any necessary fractional contribution through $s_i \in [-1, 1]$.

This is the subgradient characterization of the sample median. Therefore, in the intercept-only case, LAD estimates a median.

The bounded residual score

$$\text{sign}(r_i)$$

makes LAD less sensitive than OLS to extreme residuals in the response direction.

However, the full contribution to the LAD subgradient is

$$-x_i s_i.$$

Therefore, if x_i is extremely large, the observation may still exert substantial influence.

LAD is consequently robust to large response residuals, but it is not automatically robust to high-leverage observations in the covariates.

The LAD score in the residual direction is bounded: $\text{sign}(r)$.
A huge y does not create a huge score.

The score contribution is $x \text{sign}(r)$. A point with very large $\|x\|$ can dominate the fit.

The 50% breakdown intuition belongs to the sample median. Ordinary LAD regression can have a low breakdown point when leverage points are allowed. (Check!)

$$Q_{0.5}(Y \mid X) = X^\top \beta_0.$$
$$\hat{\beta}_{\text{LAD}} \xrightarrow{p} \beta_0.$$

The population risk $\mathbb{E}|Y - X^\top \beta|$ must have a unique minimizer at β_0 .

where

$$A = \mathbb{E}[f_{\varepsilon|X}(0 | X)XX^\top], \quad B = \frac{1}{4}\mathbb{E}(XX^\top).$$

$$\text{Avar}(\hat{\beta}_{\text{LAD}}) = \frac{1}{4f_{\varepsilon}(0)^2} Q^{-1}.$$

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The log likelihood is

$$f(\varepsilon) = \frac{1}{2b} \exp\left(-\frac{|\varepsilon|}{b}\right).$$

The log likelihood is

$$\ell(\beta, b) = -n \log(2b) - \frac{1}{b} \sum_i |y_i - x_i^\top \beta|.$$

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Efficiency depends on the error distribution



Errors	OLS	LAD
Gaussian	Optimal for the mean; $\text{ARE}(\text{LAD}, \text{OLS}) = 2/\pi \approx 0.637$	Higher asymptotic variance
Laplace	Lower efficiency (check!)	Maximum likelihood estimator and efficient
Heavy-tailed	Variance can be very large	Often stable when $f_\epsilon(0) > 0$
Contaminated	Sensitive to extreme responses	Limits the pull of vertical outliers

Relative efficiency compares asymptotic variances when both estimators target the same center under symmetric errors.

- ▶ shifting Y shifts the intercept
- ▶ multiplying Y by c multiplies the fitted coefficients and residuals by c

Then solve

$$\min_{\beta, u, v} \sum_i (u_i + v_i) \quad \text{subject to} \quad X\beta + u - v = y.$$

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	Intercept	Air flow	Water temp.	Acid conc.
OLS	-39.920	0.716	1.295	-0.152
LAD	-39.690	0.832	0.574	-0.061

The water-temperature coefficient changes substantially. Possible explanations include influential days, collinearity, model misspecification, or sampling variability.

A different coefficient is a diagnostic prompt, not automatic evidence that LAD is correct.

Source: Brownlee (1965); fits use an intercept and all three predictors.

Diabetes example: the estimands differ



Using only standardized BMI:

$$\hat{\mathbb{E}}(Y \mid \text{BMI}) = 152.13 + 949.44 \text{ BMI}$$

$$\hat{Q}_{0.5}(Y \mid \text{BMI}) = 146.34 + 1106.66 \text{ BMI}.$$

- ▶ The LAD slope describes movement in the conditional median.
- ▶ The OLS slope describes movement in the conditional mean.
- ▶ Comparing slopes is scientifically useful only after stating which center matters.

Source: Efron et al. (2004); calculations use the scikit-learn diabetes dataset.

Advantages and disadvantages relative to OLS



Dimension	LAD	OLS
Target	conditional median	conditional mean
Vertical outliers	bounded residual score	unbounded residual score
Gaussian efficiency	lower	optimal under the classical model
Heavy tails	often stable	may be unstable; variance may fail
Leverage points	still vulnerable	vulnerable
Optimization	convex but nonsmooth	closed form in basic linear models
Fit summary	MAE, pseudo- R^2	MSE, RMSE, classical R^2

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$$\min_{a,b} \sum_{i=1}^n |y_i - a - bx_i|,$$

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